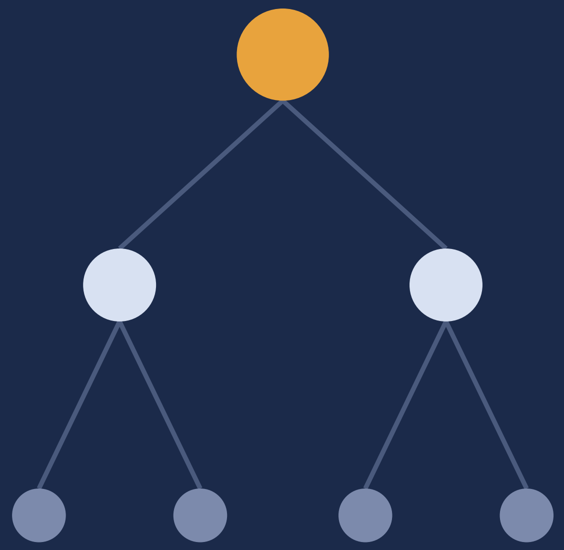
FLINS-ISKE 2026

Restricted Decision Trees

Hierarchical Feature Selection for Knowledge-Guided Staged Decision Systems



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Real Decisions Happen in Stages

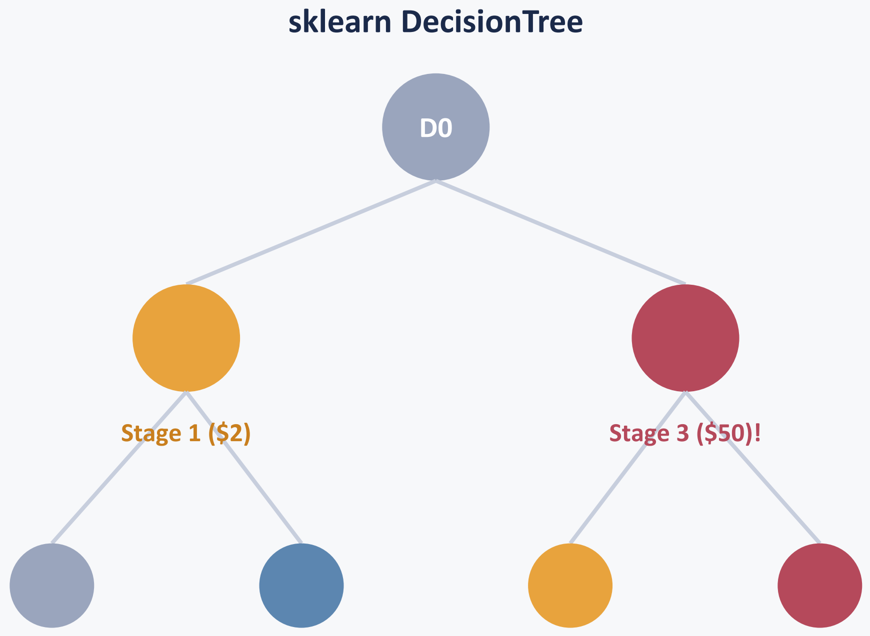
A financial institution approves or rejects a new customer by consulting sources one at a time — cheapest first, most expensive last.

STAGE 0	STAGE 1	STAGE 2	STAGE 3
$0	$1–$2	$3–$5	$10–$50
Internal Data	**Primary Verification**	**Secondary Verification**	**Enhanced Diligence**
Age, location, account history	Identity check (Vendor A)	Address validation (Vendor B)	Biometric / screening (Vendor C)

Cost ordering · Vendor specialisation · Early termination · Audit-compliant trail

Similar staged pipelines appear in medical diagnosis, fraud detection, and credit underwriting.

Standard Trees Ignore Stage Discipline



Depth 2: all four vendor stages mixed

Feature Mixing

A single node can pull from cheap and expensive vendors at once — forcing upfront acquisition of every source.

Opaque Structure

No record of which stage produced a decision, so audit and regulatory review (e.g. GDPR) can't verify compliant ordering.

No Early Stopping

Expensive sources surface at shallow depths, so obvious cases can't be resolved cheaply.

One Constraint: Same Feature Per Depth

For a tree of depth d, RDT selects features {f_0,…,f_d}. Every node at depth k splits on f_k — but each node still finds its own optimal threshold.

Same feature AND threshold per depth	Same feature, free threshold per depth	Free feature and threshold per node
ODT	**RDT**	**sklearn**
Oblivious Trees	*This Paper*	*Unrestricted*

most restricted … *unrestricted*

Direct mapping to a staged decision system

Stage k = “query the source providing feature f_k, apply the threshold test, then resolve or escalate.” Predictable cost, one vendor per stage, natural audit trail.

Note: RDT enforces one source per depth — currently operationalised as one feature per depth, not full use of a stage's available fields.

79.5% Lower Cost — No Accuracy Trade-off

Enforcing which vendor is consulted at each depth, not which of its fields are used

79.5%

projected cost reduction

on a synthetic 2,000-sample KYC simulation, with no accuracy penalty

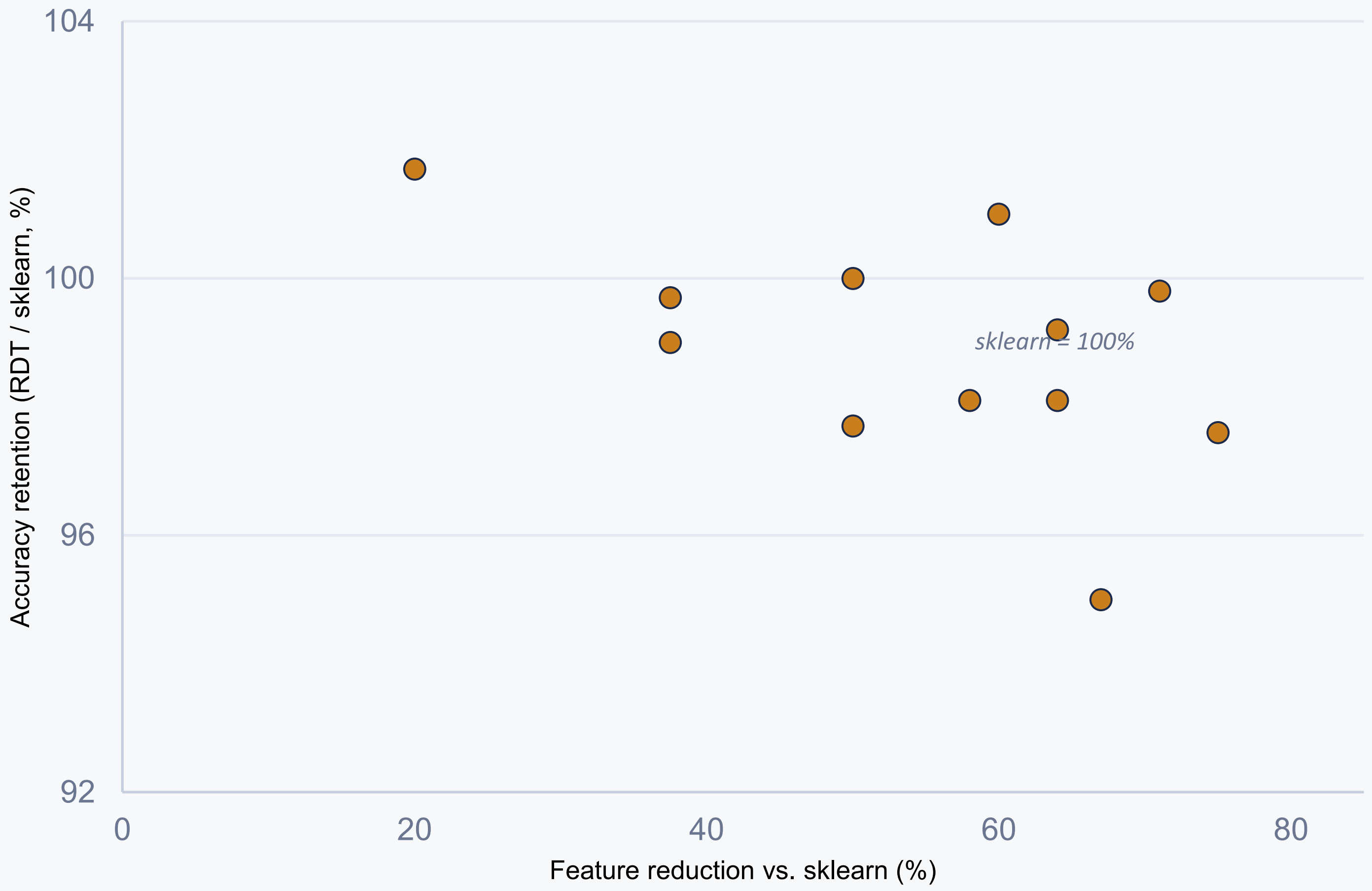
52–83%

range across industry-calibrated cost tiers & early-termination rates

Vendor stage used at each tree depth

Depth	sklearn DecisionTree	RDT
0	Stage 0 (internal, free)	Stage 0 — free
1	Stage 1 ($2) + Stage 3 ($50)	Stage 1 — $2
2	Stages 0,1,2,3 all mixed	Stage 2 — $5

99.1% of sklearn Accuracy, 58% Fewer Features



58%

average feature reduction (4.2 vs. 10.0 features)

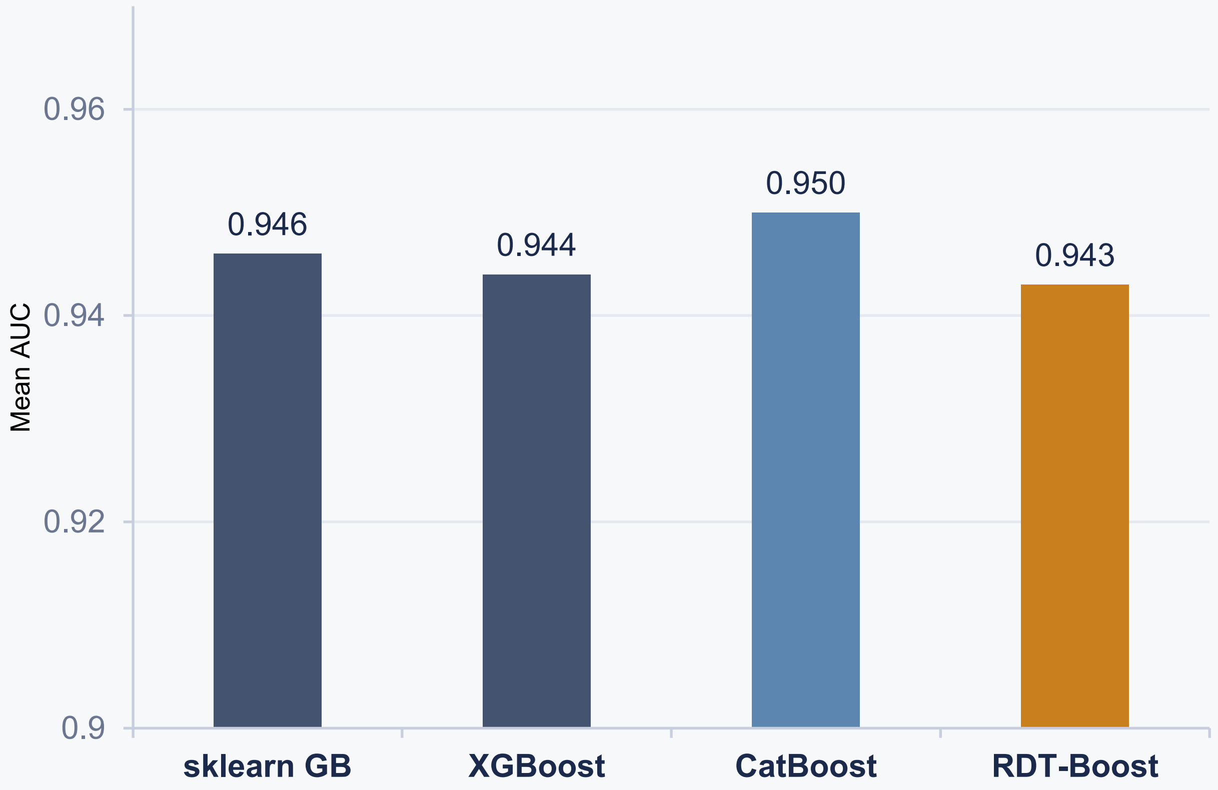
3 of 12 datasets matched or beat sklearn outright:

Adult Income — 100%

Ionosphere — 101%

Segment — 101.7%

Holds Up in Gradient Boosting Too



99.8%

of sklearn GB AUC, with 55% fewer features (379 vs. 846)

Statistically equivalent to sklearn GB

Wilcoxon n=10, W=19, p=0.432 — no significant difference

But CatBoost wins outright

p=0.002 across all 10 datasets — driven by broader feature coverage (ODT).

Why Feature Restriction Helps Ensembles

1

Reduced Greedy Overfitting

RDT-Boost uses ~2.5× fewer features per tree than sklearn GB, preventing early trees from exhausting predictive signal.

2

Enhanced Diversity

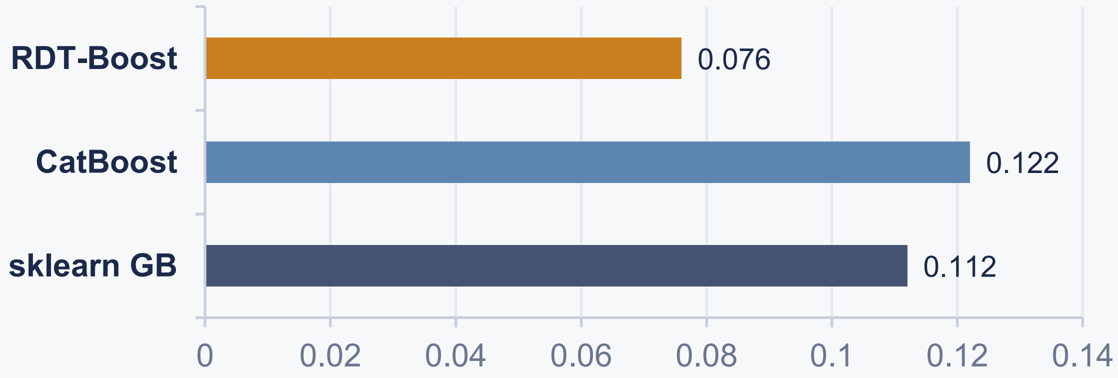
Feature restriction changes which signals each tree can see — though CatBoost's global objective still yields the most diverse ensembles.

3

Implicit Regularisation — dominant

Train–test AUC gap shrinks sharply under restriction: the clearest, most consistent effect across all datasets tested.

Train–test AUC gap on Madelon (500 features, dominant-mechanism evidence)



Lower is better — RDT-Boost shows the smallest gap between training and test AUC.

Limitations & What's Next

Limitations

- RDT enforces one vendor per depth, but currently as one feature per depth — not full use of a stage's available fields
- Feature ordering is currently a manual, domain-knowledge step — not yet cost-aware or automated
- Reference implementation is 50–100× slower than sklearn's C++ backend (Cython recovers 3–8×)
- KYC cost analysis is an illustrative synthetic simulation, not a validation on real vendor-contract data

Future Work

- Cost-sensitive RDT — fold vendor acquisition cost directly into the splitting criterion
- Block-restricted RDT — free use of a full vendor's feature subset within a depth, while still restricting each depth to one vendor
- Cost-accounted staged inference — an engine that queries each stage only for unresolved cases and bills real per-customer cost, replacing assumed termination rates with measured ones
- PAC-learning bounds for RDT's constrained hypothesis class
- Full Cython / C++ implementation for sklearn-comparable runtime

One Constraint, Structural Benefits

99.1% / 99.8% — of sklearn / sklearn-GB performance retained — standalone and boosted

55–58% — fewer features used, across 12 standalone + 10 boosting datasets

$p = 0.432$ — RDT-Boost's AUC is statistically indistinguishable from sklearn GB across 10 datasets (Wilcoxon signed-rank) — in an ensemble, the restriction costs nothing measurable

Thank you — questions welcome.

Code and reproduction: github.com/amritnm/RDT-Reproduction-FLINS-ISKE-2026

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